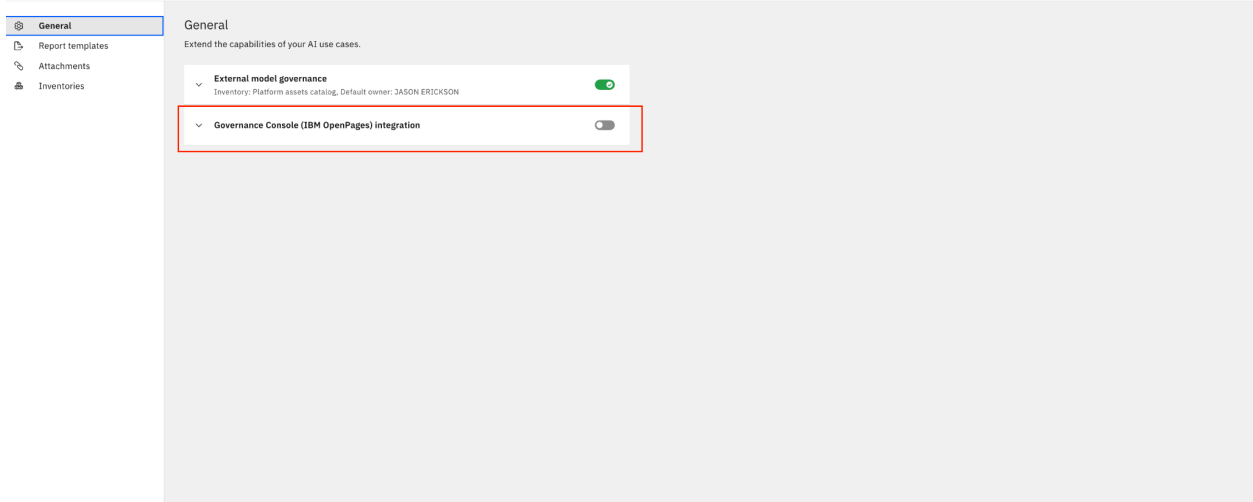


Lab 7 Use case creation and AI Factsheets

Spoiler: For the purpose of this exercise we will not do an integration with IBM Open Page, a workflow and risk management tool for AI risk manager and business users. Instead we will focus more on the lower level technical AI fact sheets and basic usecase creation. Please request an End-to-end demo for the full experience.

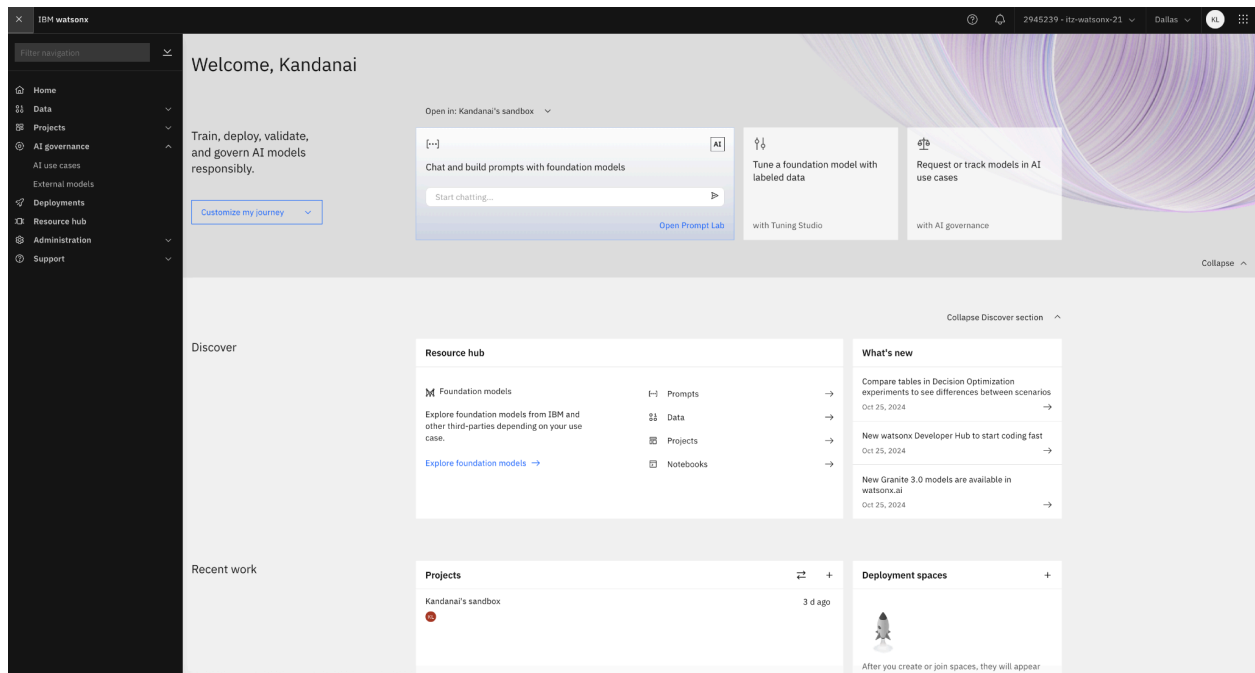
Manage

Control general settings of the AI use cases available to you.



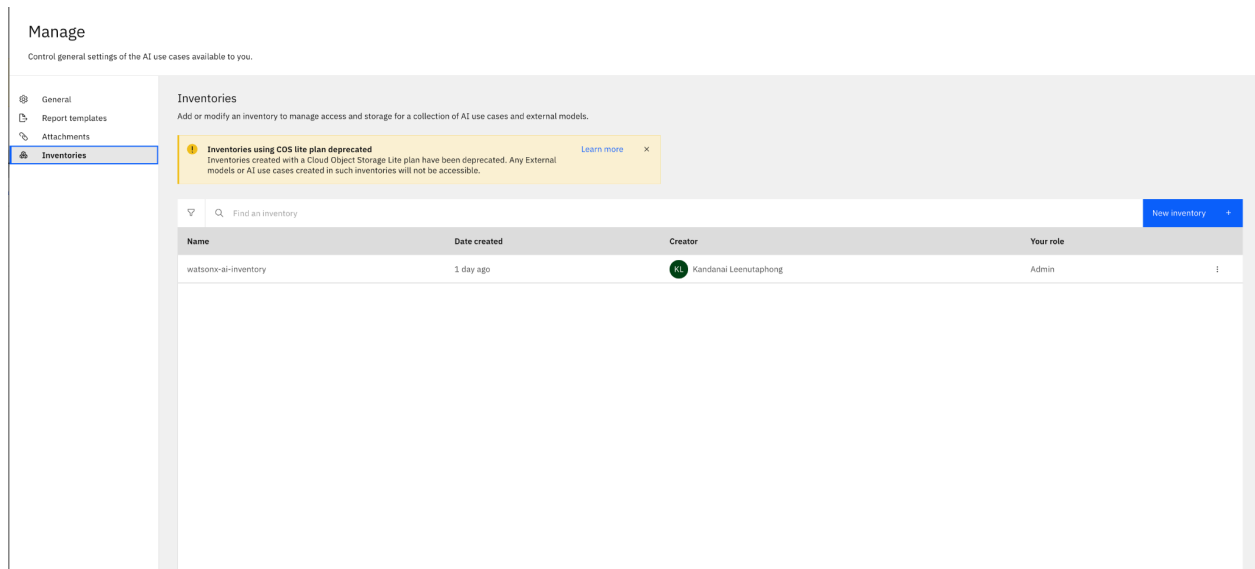
Learning objectives:

Step 1: The first step you will need to do is to create an AI Usecase. To do so, please go to “AI Governance” then AI usecases.



Step 2: Before you can create an AI Usecase you must create an inventory. Inventory can be found in the settings icon inside AI usecase. Below shows an example of a created inventory. If you have not already gotten an inventory please create one.

Definition/purpose of inventory:



New Inventory

Name

watsonx-ai-inventory

Description (optional)

Description text

☒ Add collaborators after creation

An inventory without collaborators will be visible to only you. Add members to collaborate on AI use cases.

IBM Cloud Object Storage ⓘ

This service stores the files associated with assets in the inventory.

Object storage instance

it2cos-6910005csg-yips8buw

Or create a new Cloud Object Storage instance ⓘ

Cancel

Submitting...

Step 3: After creating a new inventory you can now create an AI usecase.

AI use case ⓘ

▽

Find a AI use case

New AI use case

Name	Status	Owner	Inventory/Catalog	Tags	Risk level	Alerts in
 No AI use case available Create a use case to define a business problem, request a model, and specify details such as deadlines or requirements. New AI use case +						

Items per page: 20 ▾

0-0 of 0 items

Page 1 < >

New AI use case

Create a use case to define a business problem, request a model, and specify details such as risk level and status.

General information

Name (Required)

Description

Risk level

Medium

Custom risk label

Inventory/Catalog (Required)

Details

Purpose

What is the purpose of this AI use case?

Supporting documentation

Please enter a URL pointing to external documentation

Owner

KL

Kandanae Leenutaphong

Kandanae.Leenutaphong@ibm.com

Status

Ready for use case approval

Tags

Search tags

Add tags to this AI use case.

Cancel

Create

Here please add information such as the name of the use case, the basic risk level associated with having this use case, and the associated inventory. Be sure to add the purpose, description or any supporting documentation.

Step 4: Here we can see that there will be unassociated workspaces from development, validation (testing) to operation (production). In order to associate these workspaces, we will need to create an asset in the project, and two deployment spaces (testing, production).

Overview

Lifecycle

Access

Welcome to your new AI use case

Discover next steps

Get your AI use case status

Check the status of your AI use case. If it is in a pending state, you can click on the status icon to see the details.

Get started with samples

See the samples and get started with action with a sample project. You can also find the samples in the project page.

General information

Name

Loan approval model

Description

some description

Owner

Kandanae Leenutaphong

Kandanae.Leenutaphong@ibm.com

Tags

Add tags to this AI use case.

Details

Purpose

What is the purpose of this AI use case?

Associated workspaces

Associate your AI use case with the workspaces in order to organize them under the same business problem.

Development

Associate workspaces

Validation

Associate workspaces

Operation

Associate workspaces

Attachments

Other attachments

Drag and drop files here or click to upload

About this AI use case

Name

Loan approval model

Description

some description

Owner

Kandanae Leenutaphong

Kandanae.Leenutaphong@ibm.com

Status

Ready for use case approval (Kandanae.Leenutaphong, Oct 27, 2024)

Risk level

Medium

Inventory/Catalog

watsonx-ai-inventory

Tags

Add tags to this AI use case.

Created by

Kandanae Leenutaphong, Oct 27, 2024

Modified by

Kandanae Leenutaphong, Oct 27, 2024

Step 5: Create a project as you have done in the previous lab. Here I named the project as “Development-project” but you would have other meaning. Next, upload the csv file from lab1 and repeat Lab 2 exercises of running the Auto AI experiment. But, Do not deploy/promote the model to deployment space yet.

	Name	Last modified
☐	loan_exp - F2 Snap Random Forest Classifier - Model Machine Learning model from AutoML	Now Modified by Service
☐	loan_exp AutoAI experiment	1 minute ago Modified by you
☐	dataset_loan_example_v2.csv CSV	7 minutes ago Modified by you

Step 6: Here, as a next step please return to the use case. You can now associate the development stage with your newly created project.

Development

To this phase your development team will experiment with different approaches of ML development or prompt engineering and test them in projects.

[Associate workspace](#)

Validates

To this phase your validation team will evaluate models in pre-production deployment spaces and prompt templates in projects.

[Associate workspace](#)

Operation

To this phase your development team will evaluate models in pre-production deployment spaces.

[Associate workspace](#)

Attachments

Other attachments

Drag and drop files here or click to upload

About this AI use case

Name	Description	Owner	Status	Risk level	Inventory/Catalog	Supporting documentation
Loan approval model	Loan approval model	Yaroslav Lermontov	Ready for use case approval (Operational Learning, Oct 27, 2024)	Medium	Inventory/Catalog	Placeholder for external documentation

Diagram: A circular flow diagram showing the stages of an AI use case: Development for Review, Deployment for Review, and Deployment for Production.

Step 7: In order to track the model what needs to be done is please go to your project and open the Model asset saved from AUTO AI. You should see that the asset is not tracked. Please press on track AI usecase.

The screenshot shows the 'Governance' page for a model asset named 'loan_exp - P2 Snap Random Forest Classifier - Model'. The page indicates that the asset is not tracked and provides a 'Track to AI use case' button. Below this, the 'Model' section displays metadata including creation and modification dates, the approach used ('SnapRandomForestClassifier'), and the production type ('snap-hybrid_0.1'). The 'Training: General' section shows the project name and a link to the training configuration. The 'Input schema' section lists various features and their types.

Asset Details:

- Name: loan_exp - P2 Snap Random Forest Classifier - Model
- Description: No description provided.
- Asset Details:
 - Type: snap-hybrid_0.1
 - Model ID: snap-hybrid_0.1-0.1
 - Software configuration: snap-hybrid_0.1-0.1
 - Model (please software specifications): snap-hybrid_0.1-0.1
- Tags: Add tags to make assets easier to find.
- Last modified: 28 seconds ago by Service
- Created on: Oct 28, 2024 by Randeev Lennathapong

Model Metadata:

Field	Value
Created by	Randeev Lennathapong
Created	October 28, 2024 at 08:32:02 AM
Modified	October 28, 2024 at 08:32:18 AM
Approach	SnapRandomForestClassifier
Model type	snap-hybrid_0.1
Production type	classification

Training: General

Project: Development-Project

Link: [autoai-h24-3-p2-0.1.1](#)

Input schema

Feature	Type
AccountBalance	Integer
CreditScore	Boolean
CreditHistory	Boolean
Dependents	Boolean
Education	Other
Gender	Other
LoanAmount_Term	Boolean
LoanAmount	Boolean
Married	Other
PropertyType	Other
Self_Employed	Other

Step 8: Here you can select the default approach (experimental template).

The screenshot shows the 'Track asset' dialog box. It has a sidebar with three options: 'Define approach' (selected), 'Assign version', and 'Review'. The main area is titled 'Define approach' and contains a 'Use case: Loan approval model' section. Below this, it states: 'An approach defines one path for solving the goal of the use case. Ex. an approach might be a variation on a machine learning model, or a challenger model. Each approach can include multiple versions.' There is a 'New approach +' button. A card titled 'Default approach' is displayed, stating: 'A default approach for tracking your AI assets.' At the bottom, there are 'Cancel', 'Back', and 'Next' buttons.

Track asset

Track a model and related assets through the AI lifecycle, from request to productive use.

Define approach

Assign version

Review

Assign version

Approach: **Default approach** | Use case: **Loan approval model**
Choose the starting point for this approach.



Experimental

Use this as a starting point if your model is just starting in development and its input and output structure will likely change in the near future.

0.0.1



Stable

Use this as a starting point if your model is in a production state and you won't expect any major changes in its input and output structure soon.

1.0.0



Custom

Define your own starting version if you already tracked this model in a versioning context before.

Version number

0.0.1

Comment (optional)

Provide context for this change. Comments are saved in the model submit to govern.

Cancel

Back

Next

Track asset

Track a model and related assets through the AI lifecycle, from request to productive use.

Define approach

Assign version

Review

Review

Loan approval model

Approach



Default approach

Version



0.0.1

A default approach for tracking your AI assets.

00800080-0000-0000-0000-000000000000

Cancel

Back

Track asset

It should show in your AI fact sheet that the use case is now tracked. In this fact sheet you could also see all the experimental results, further information like the model used, inference endpoint (if you had deployed the model) and extra information like evaluations all in a fact sheet.

The screenshot displays the IBM Watson AI Governance interface. The left sidebar shows a navigation menu with 'Governance' selected, containing 'Model', 'Development', and 'Development-Project'. The main content area is titled 'Loan approval model' and includes a description, approach (Default approach), version (0.0.1), and a lifecycle diagram with stages: Develop, Validate, and Operate. The right sidebar shows 'About this asset' with details like Name, Description, Asset Details, and Tags. The bottom section shows the 'Model' details for 'loan_exp - P2 Snap Random Forest Classifier - Model'.

Step 9: If you head back to the loan approval use case and click on the life cycle. You should see the asset and the project being associated as in the image below.

The screenshot displays the IBM Watson AI Governance interface, specifically the 'Loan approval model' lifecycle view. The left sidebar shows a navigation menu with 'Overview' and 'Lifecycle' selected. The main content area is titled 'Loan approval model' and includes a description, approach (Default approach), version (0.0.1), and a lifecycle diagram with stages: Develop, Validate, and Operate. The right sidebar shows 'About this AI use case' with details like Name, Description, Asset Details, and Tags. The bottom section shows the 'Model' details for 'loan_exp - P2 Snap Random Forest Classifier - Model'.

AI use cases / Loan approval model

Overview **Lifecycle** Access

Welcome to your new AI use case

Discover next steps

Organize team work with approaches

There can be different paths to solving a business problem. Use an approach to organize the assets for one solution. Name and define the approach, then invite team members to track assets in them.

Manage AI assets with versions

Each approach starts a new version series for identifying the associated assets. You can preserve a particular state of assets in one approach, then create a new approach for exploring a new direction.

Show deleted assets New approach +

Default approach

Latest: 0.0.1 | 1 version

A default approach for tracking your AI assets.

Development
Experiment with different approaches of ML development or prompt engineering and test them in projects.

Validation
Evaluate models and prompt templates with validation data.

Operation
Evaluate and monitor models and prompt templates in production.

v0.0.1

Development-Project

loan_exp - P2 Snap Random Forest Classifier - Model

No models promoted to a pre-production space.

No models promoted to a production space.

Step 10: As a next step we can assume that the technical team ie. (Data Scientist) is satisfied with the result. As a next step, we will create two deployment spaces (one for Testing & and one for production). Of which, we can let the QA/QC team test the deployed model before we deploy the model to production.

Creating testing deployment space:

Create a deployment space

Use a space to collect assets in one place to create, run, and manage deployments

Define details

Name

Testing-deployment-space-loan-approval

Description (Optional)

Deployment space description

Deployment stage

Testing

Deployment space tags (optional)

Find or create tags

Select services

Select storage service

itccos-4910055cgg-yiqs8buv

Select machine learning service (optional)

itxmi-6910005cgg-yiqs8buv

Upload space assets (optional)

Populate your space with assets exported from a project or space to a .zip file. You can add more assets after the space is created.

Drop .zip file here or browse your files to upload

Cancel

Create

Creating a production deployment space:

Create a deployment space

Use a space to collect assets in one place to create, run, and manage deployments

Define details

Name

Production-deployment-space-loan-approval

Description (Optional)

Deployment space description

Deployment stage

Production

Deployment space tags (optional)

Find or create tags

Select services

Select storage service

ltzcos-691005ccg-yiqs8buw

Select machine learning service (optional)

Select a machine learning service

Upload space assets (optional)

Populate your space with assets exported from a project or space to a .zip file. You can add more assets after the space is created.

Drop .zip file here or browse your files to upload

Cancel

Create

Once created you should see the deployment spaces in the overview page of watsonx.

Collapse Discover section

Resource hub

Foundation models

Explore foundation models from IBM and other third-parties depending on your use case.

Explore foundation models →

Prompts

Data

Projects

Notebooks

What's new

Compare tables in Decision Optimization experiments to see differences between scenarios

Oct 25, 2024

→

New watsonx Developer Hub to start coding fast

Oct 25, 2024

→

New Granite 3.0 models are available in watsonx.ai

Oct 25, 2024

→

Projects

Development-Project

32 min ago

Kandanai's sandbox

4 d ago

Deployment spaces

Production-deployment-space-loan-approval

7 min ago

Testing-deployment-space-loan-approval

8 min ago

Step 11: Here we will be promoting the model asset from AUTO AI to the testing space and deploy the model.

Find assets

Import assets

New asset

3 assets

All assets

Asset types

Data

Experiments

Models

Name	Last modified	
loan_exp - P2 Snap Random Forest Classifier - Model	21 minutes ago Modified by Service	
loan_exp	29 minutes ago Modified by you	Promote to space
dataset_loan_example_v2.csv	35 minutes ago Modified by you	Delete

Promote to space

Promote the asset to a deployment space to deploy the asset or to support a deployment.

Target space

Testing-deployment-space-loan-approval

Create a new deployment space

Testing-deployment-space-loan-approval

Production-deployment-space-loan-approval

Testing-deployment-space-loan-approval

Description of assets

Find or create tags

Promoting a version of an asset to a space creates a new asset in the space, with a new asset ID.

Selected assets (1)

Name	Format	Version	Status
loan_exp - P2 Snap Random Forest ...	Model	Current	Queued

Cancel

Promote

Deployments AI Factsheet

Search

New deployment

Name	Type	Status	Tags	Last modified
loan-qa-qc	Online	Deployed		26 seconds ago Kandana Leenutaphong (You)

Step 12: Once again please go back to the AI usecase to associate the testing deployment space with the Loan approval AI usecase

The screenshot displays the IBM watsonx AI use case management interface. The top navigation bar includes 'Home', 'Data', 'Projects', 'AI governance', 'AI use cases', 'External model', 'Deployments', 'Resource hub', 'Administration', and 'Support'. The 'AI use cases' section is highlighted.

The main content area shows the 'Loan approval model' use case. The 'General information' section includes:

- Name:** Loan approval model
- Description:** some description
- Owner:** Kandana Leenutaphong (kandana.leenutaphong@ibm.com)
- Status:** Ready for use case approval (Kandana Leenutaphong, Oct 27, 2024)
- Risk level:** Medium
- Inventory/Catalog:** watsonx-ai-inventory

The 'Details' section includes:

- Purpose:** What is the purpose of this AI use case?
- Supporting documentation:** Please enter a URL pointing to external documentation

The 'Associated workspaces' section shows the following workspaces:

- Development:** Development-Project
- Validation:** Associate workspaces +
- Operation:** Associate workspaces +

The 'About this AI use case' sidebar on the right provides additional details:

- Name:** Loan approval model
- Description:** some description
- Owner:** Kandana Leenutaphong (kandana.leenutaphong@ibm.com)
- Status:** Ready for use case approval (Kandana Leenutaphong, Oct 27, 2024)
- Risk level:** Medium
- Inventory/Catalog:** watsonx-ai-inventory
- Tags:** Add tags to this AI use case.
- Created by:** Kandana Leenutaphong, Oct 27, 2024
- Modified by:** System, Oct 28, 2024

A diagram on the right illustrates the AI use case lifecycle, showing the flow from 'Workspaces for Development' to 'Workspaces for Operate' and 'Workspaces for Validate'.

10/14 Watsonx

[Deployments](#) / [Testing-deployment-space-loan-approval](#) / [loan_exp - P2 Snap Random Forest Classifier - Model](#)

Deployments

AI Factsheet

Governance

Model

Development

Validation

Testing-deployment...

Training evaluation

Additional details

Attachments

Governance

This model was automatically untracked due to an association conflict.

A space can only be associated to one phase of one AI use case. Importing this asset would've caused a conflict and as a result, your asset has been untracked automatically. You can track it again based on the association of this workspace.

This asset is not tracked.

To track an asset, add it to an AI use case. Tracking captures details about the asset for governance purposes.

[Track in AI use case](#)

Model

loan_exp - P2 Snap Random Forest Classifier - Model

086df54b-9b01-4aa3-39ef-b0f8222599a0

Created by

Kandanai Leenutaphong

Created

October 28, 2024 at 09:01:03 AM

Modified

October 28, 2024 at 09:01:05 AM

Algorithm

SnapRandomForestClassifier

Model type

wml-hybrid_0.1

Software specification

hybrid_0.1

Label column

Y

Prediction type

classification

Training: General

About this asset

Name

loan_exp - P2 Snap Random Forest Classifier - Model

Description

No description provided.

Asset Details

Type: wml-hybrid_0.1

Model ID: 086df54b-9b01-4aa3-39ef-b0f8222599a0

Software specification: hybrid_0.1

Hybrid pipeline software specifications: wml-hybrid_0.1

Tags

Add tags to make assets easier to find.

Source asset details

Last modified

4 minutes ago by Service

Created on

Oct 28, 2024 by Kandanai Leenutaphong

loan_exp - P2 Snap Random Forest Classifier - Model

Track asset

Track a model and related assets through the AI lifecycle, from request to productive use.

Define AI use case

Define approach

Assign version

Review

Define AI use case

Choose which associated AI use case this asset should be tracked under.

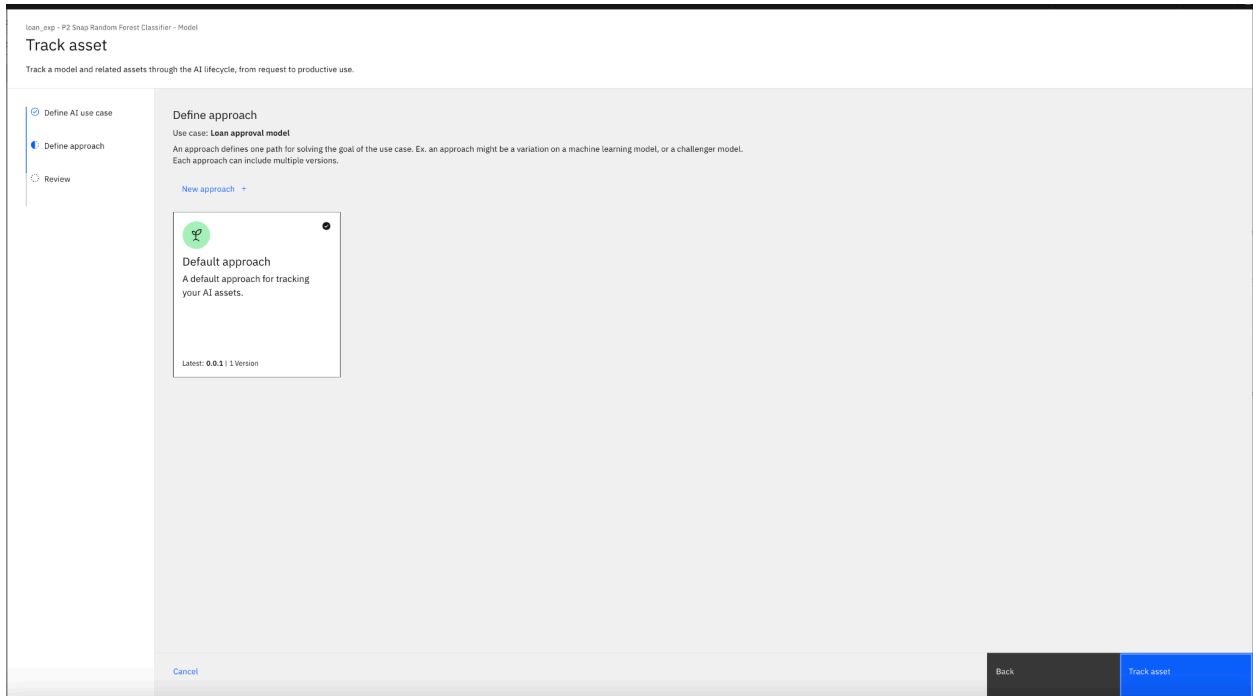
Find AI use cases

Name	Inventory	Status	Owner
Loan approval model	watson-ai-inventory	Ready for use case approval	
Description No description defined			

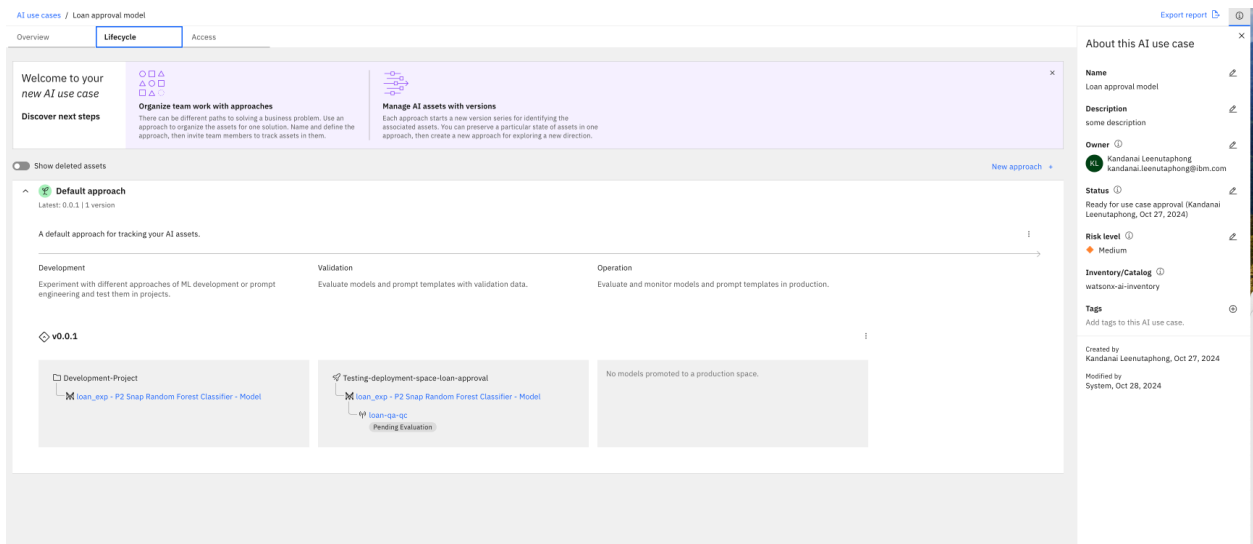
Cancel

Back

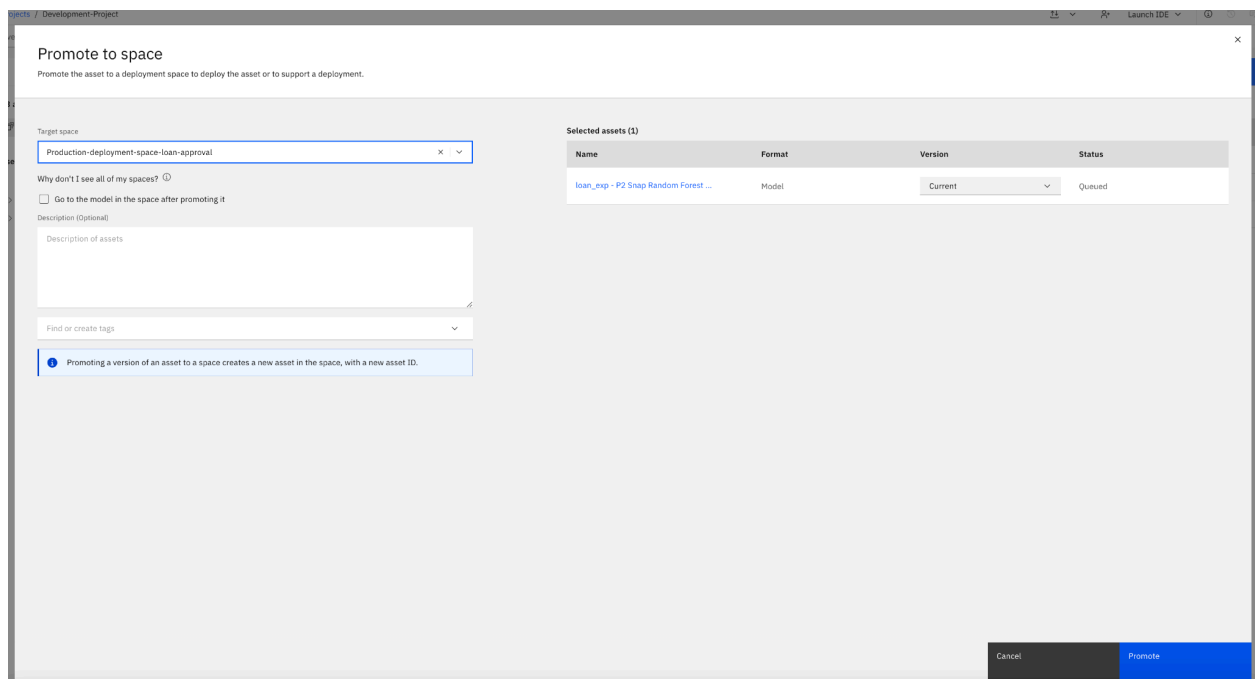
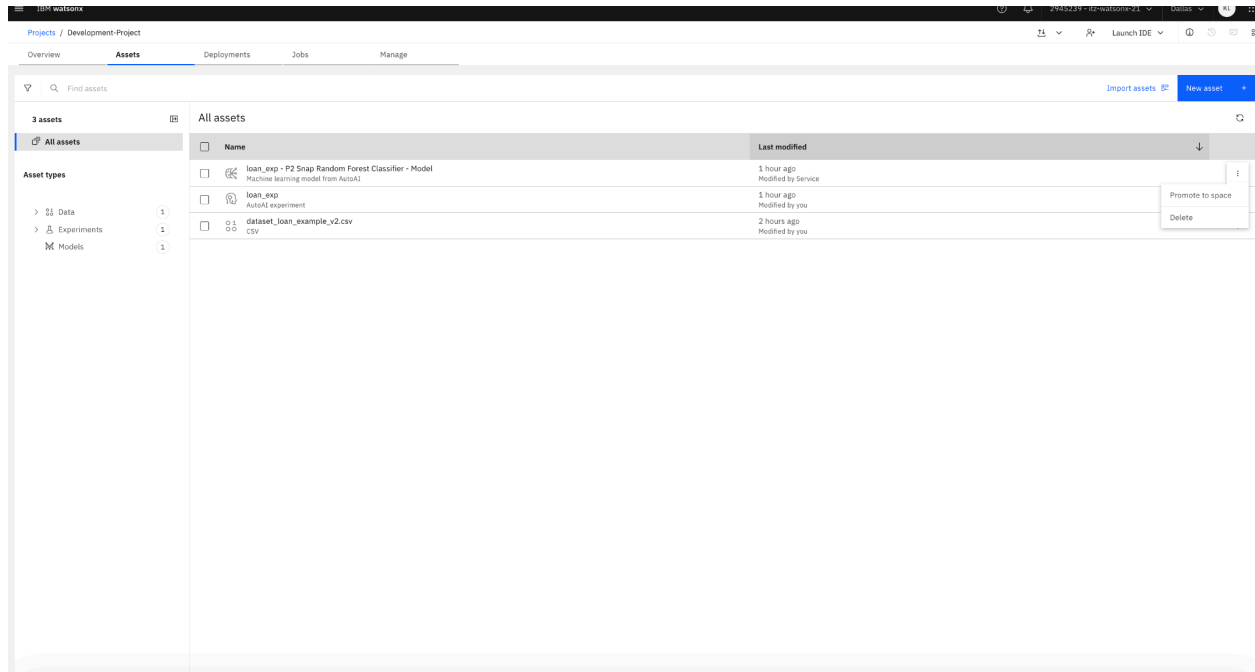
Next



Step 14: Once tracked, you should see that the model for qa/qc testing will be tracked in the validation phase. Ready for the evaluation. Here the QA/QC team could test if the model is performing well and if there are any metrics breaching the threshold. If there are any need for further improvement, we could create another experimental model version and go back to development (But, we will skip that for this exercise)



Step 15: Here we will Repeat the steps of promoting the model from the development project to the production deployment space and deploying the model.



IBM watsonx

Deployments / Production-deployment-space-loan-approval / loan_exp - P2 Snap Random Forest Classifier - Model

Deployments AI Factsheet

Search

New deployment

Name	Type	Status	Tags	Last modified
proddeployan	Online	Deployed		20 seconds ago Kandana Leenutaphong (You)

Items per page: 20 1-1 of 1 items 1 of 1 pages

About this asset

Name
loan_exp - P2 Snap Random Forest Classifier - Model

Description
No description provided.

Asset Details
Type: em-hybrid_0.1
Model ID: 38fdec77-441a-44...
Software specification:
hybrid_0.1 @
Hybrid pipeline software specifications:
autoai-kb_r24.1-py3.11

Tags
Add tags to make assets easier to find.

Source asset details

Last modified
1 minute ago by Service
Created on
Oct 28, 2024 by Kandana Leenutaphong

Step 16: We would again, repeat the step of associating the production deployment space with the AI Use case

Loan approval model

Associate workspaces Operation

How it works

Association rules
For each AI use case, associate workspaces with lifecycle phases to organize assets for a business problem. Associate a project to develop assets, a project or space to validate assets, and an operate space for assets in production.
[Read more](#)

Why is this important?
Organize your associations before you start tracking assets in an AI use case. Once asset tracking is enabled, the associated workspace cannot be updated without losing the tracked data.
[Read more](#)

Space
A space can be associated with multiple AI use cases.

Find spaces New space +

Name	Created	Your role	Tracked AI assets	Stage	Associates
☒ Production-deployment-space-loan-approval	10/28/2024	Admin	0	Production	0
☐ Testing-deployment-space-loan-approval	10/28/2024	Admin	1	Testing	1

Cancel Save

Overview

Lifecycle

Access

CONNECTION FOR ALL PARTICIPANTS IN THIS USE CASE

View sample project →

General information

Name

Loan approval model

Description

some description

Owner

Kandana Leenutaphong

kandana.leenutaphong@ibm.com

Tags

Add tags to this AI use case.

Status

Ready for use case approval (Kandana Leenutaphong, Oct 27, 2024)

Risk level

Medium

Inventory/Catalog

watsonx-ai-inventory

Details

Purpose

What's the purpose of this AI use case?

Supporting documentation

Please enter a URL pointing to external documentation

Associated workspaces

Associate your AI use case with the workspaces in order to organise them **under the same business problem.**

Development

Development-Project

Validation

Testing-deployment-space-loan-approval

Operation

Production-deployment-space-loan-approval

About this AI use case

Name

Loan approval model

Description

some description

Owner

Kandana Leenutaphong

kandana.leenutaphong@ibm.com

Status

Ready for use case approval (Kandana Leenutaphong, Oct 27, 2024)

Risk level

Medium

Inventory/Catalog

watsonx-ai-inventory

Tags

Add tags to this AI use case.

Created by

Kandana Leenutaphong, Oct 27, 2024

Modified by

System, Oct 28, 2024

Step 17: Once again, please track the AI use case

Deployments / Production-deployment-space-loan-approval / loan_exp - P2 Snap Random Forest Classifier - Model

Deployments

AI Factsheet

Governance

Model

Unspecified phase

Training evaluation

Production-deployment...

Governance

This model was automatically untracked due to an association conflict.

A space can only be associated to one phase of one AI use case. Importing this asset would've caused a conflict and as a result, your asset has been untracked automatically. You can track it again based on the association of this workspace.

This asset is not tracked.

To track an asset, add it to an AI use case. Tracking captures details about the asset for governance purposes.

Track in AI use case

Model

loan_exp - P2 Snap Random Forest Classifier - Model

Track asset

Track a model and related assets through the AI lifecycle, from request to productive use.

- Define AI use case
- Define approach
- Assign version
- Review

Define AI use case

Choose which associated AI use case this asset should be tracked under.

Find AI use cases

Name	Inventory	Status	Owner
Loan approval model	watsonx-ai-inventory	Ready for use case approval	

Cancel

Back

Next

Deployments

AI Factsheet

Governance

Model

Development

Development-Project

Training evaluation

Additional details

Attachments

Validation

Testing-deployment...

loan-qa-qc

Additional details

Attachments

Operation

Production-deploy...

proddeployan

Additional details

Attachments

Governance

Export report

Loan approval model

Ready for use case approval | Kandanai Leenutaphong | watsonx-ai-inventory | medium | 81b1c15e-6a3f-4f38-8a2e-efcf55948f07

Description

some description

Approach

Default approach

Version

0.0.1

A default approach for tracking your AI assets.

00000000-0000-0000-0000-000000000000

Lifecycle

01

Develop

02

Validate

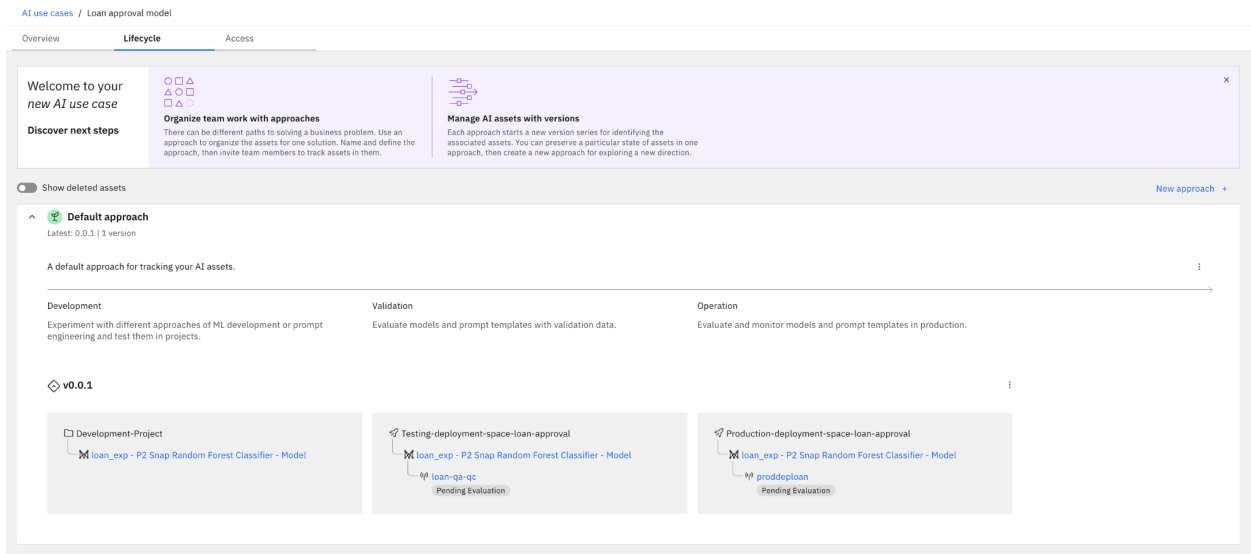
03

Operate

Untrack

Model

Finally, you will see the model being tracked across developing, validation and production. Let's say there is an issue with the model in the future, the technical team can create another model version and go through the “development”, “validation” and “Operation” stage again.



you will get to see the AI use case getting moved from ready for use case approval to Deployed. Here it is done by clicking on the “status” of the use case. With Openpage options enabled, this experience will be even more seamless.

